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Professional Preparation

December 2014 Ph.D., Computer Science, Yale University
Dissertation: *Robotic Self-Modeling*
Committee: Brian Scassellati (Advisor), Steven W. Zucker
Aaron Dollar (Mechanical Engineering), Chad Jenkins (Brown University)
May 2010 M.Phil., Computer Science, Yale University
May 2008 M.S., Computer Science, Yale University
January 2006 M.Eng., Computer Science, Cornell University
May 2001 B.S., Computer Science (Cum Laude), West Virginia University

Appointments

03/19 - 9/22	Assistant Director	Texas Robotics University of Texas at Austin
09/17 -	Assistant Professor of Practice	Department of Computer Science University of Texas at Austin
12/16 - 09/17	Postdoctoral Fellow	Department of Computer Science University of Texas at Austin
11/14 - 12/16	Postdoctoral Fellow	Department of Mechanical Engineering University of British Columbia
09/13 - 11/14	Visiting Scholar	Department of Mechanical Engineering University of British Columbia
09/06 - 12/14	Ph.D. Candidate	Department of Computer Science Yale University
06/05 - 09/06	Research Assistant	Intelligent Information Systems Institute Cornell University
12/01 - 12/04	Software Engineer	SFA, Incorporated

Teaching

- CS 309 - Autonomous Intelligent Robotics - FRI I: Spring 2018-present (Spring Semesters)
 - This course prepares students for participation in robotics research in my & Peter Stone's laboratory. Students learn C++ programming, Robot Operating System (ROS), basic OpenCV programming, and how to compute spatial transformations and use the ROS TF service. The class concludes with a final project chosen by the students and programmed on our laboratory's real robots.

- CS 378 - Autonomous Intelligent Robotics - FRI II: Fall 2017-present (Spring Semesters)
 - This is a research-based course in which students carry out semester-long research projects as teams. In 2019, I entirely reworked the class to be centered around a simulation of the graduate school process. Students submit a project proposal at the start of the class, a “workshop paper” for the midterm, and a “conference paper” for the final. The class features a peer-review process, a “program committee meeting,” a “workshop talk,” and a “conference talk.” Projects from this class have gone on to be published in top-tier conferences. I am very proud of how good this class is.

Other Professional Activities

JOURNAL EDITORIAL BOARDS

- Associate Editor - ACM Transactions on Human-Robot Interaction: 2021-current (current as of 2026)
- Associate Editor - Frontiers in Robotics and AI: 2020-2025

CONFERENCE ORGANIZING

- Organizing Committee - IEEE International Conference on Robot-Human Interactive Communication (RO-MAN; Late Breaking Report Co-Chair). 2028 (bidding starts 2 years in advance)
- Organizing Committee - ACM/IEEE International Conference on Human-Robot Interaction (HRI; Video and Demo Track Co-Chair). 2027 (bidding starts 2 years in advance)
- General Co-Chair - Texas Symposium on Machine Learning, Responsible AI & Robotics. March 3–4, 2026.
- Organizing Committee - Workshop on Benchmarking Autonomous Service Robotics in Real Life, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). Abu Dhabi, United Arab Emirates. October 15, 2024.
- Organizing Committee - Forum on Enhancing Service Robots for Future Homes held at the IEEE/RAS International Conference on Robotics and Automation. 2024
- General Co-Chair - International Symposium on Trustworthy Autonomous Systems. 2024
- Organizing Committee - International Symposium on Technical Advances in Human-Robot Interaction. 2024
- Organizing Committee - IEEE International Conference on Development and Learning (ICDL). To be hosted at UT Austin in 2024.
- Publicity Committee - International Conference on Social Robotics (ICSR) 2023. Co-Chair.
- Organizers - Forum on Enhancing Service Robots for Future Homes: Insights and Learnings at ICRA: 2024 (Pending Approval)
- Organizing / Program Committee - AAAI Fall Symposium on Artificial Intelligence for Human-Robot Interaction (AI-HRI): 2019-2023. General Co-Chair in 2019.¹. Program Committee Chair in 2020. Program Committee Co-Chair in 2023.

¹With Richard Freedman.

- Organizing / Program Committee - AAAI Spring Symposium on Machine Learning for Mobile Robot Navigation in the Wild. March 22-24, 2021
- Organizing / Program Committee - HRI Workshop on Human Interactive Robot Learning. March 7, 2022 & March 12, 2023
- Organizing / Program Committee Co-Chair - HRI Workshop on Longitudinal Social Impacts of HRI over Long-Term Deployments. March 11, 2022
- Organizing / Program Committee Chair - HRI Workshop on Exploring Applications for Autonomous Nonverbal Human-Robot Interaction.
- Executive Committee RoboCup@Home: 2019-present²
- Technical Committee RoboCup@Home: 2018

PROGRAM COMMITTEE MEMBERSHIP

- Associate Editor - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2026
- Associate Editor - IEEE International Conference on Development and Learning (ICDL): 2025
- Area Chair - Conference on Robot Learning (CoRL): 2026
- Association for the Advancement of Artificial Intelligence (AAAI): 2018-2024
- International Joint Conference on Artificial Intelligence (IJCAI): IJCAI 2011, 2020, Senior Program Committee - 2021, Novel Program Committee (3 year term) 2022-2024
- Program Committee Member - ACM/IEEE International Conference on Human-Robot Interaction (HRI; PC membership is especially competitive for HRI): 2022 (Technical Advances), 2024-2025 (Systems).
- Area Chair - ACM/IEEE International Conference on Human-Robot Interaction (HRI), Systems: 2026.
- International Conference on Educational Advances in Artificial Intelligence (EAAI): 2022
- International Conference on Autonomous Agents and Multiagent Systems: 2019
- IEEE International Conference on Robotics and Automation (ICRA) - Associate Editor: 2023 (2024 conference)

JOURNAL REVIEWER

- Frontiers in Robotics and AI: 2026
- Springer Artificial Intelligence Review: 2024
- Robotics and Automation Letters (RA-L): 2022³

²Split into 3-year terms.

³Tracked for presentation at IEEE RAS/EMBS International Conference on Biomedical Robotics & Biomechanics (BioRob)

- Robotics and Automation Letters (RA-L): 2021, 2022, 2023, 2026
- Springer Artificial Intelligence Review: 2021
- ACM Transactions on Applied Perception (TAP): 2017
- IEEE Transactions on Human-Machine Systems (THMS): 2017
- The International Journal of Human-Computer Studies (IJHCS): 2016-2018, 2022
- The International Journal of Humanoid Robotics (IJHR): 2009-2019, 2021
- Pattern Recognition letters: 2016
- ACM Transactions on Interactive Intelligent Systems (TIIS): 2014
- ASME Journal of Dynamic Systems, Measurement and Control: 2015
- Robotics and Computer Integrated Manufacturing: 2015
- Journal of Intelligent and Robotic Systems (JINT) Special Issue - Cognitive Robotics Systems: Concepts and Applications: 2014-2015
- International Journal of Robotics Research (IJRR): 2015, 2017
- Journal of Human-Robot Interaction (JHRI): 2012 (reviewed for the inaugural issue)
- International Journal of Social Robotics (SORO): 2013, 2012, 2010
- Science: Robotics: 2022
- Transactions on Human-Robot Interaction: 2018-2021 (Became an AE in 2021)
- IEEE Transactions on Human-Machine Systems: 2017
- International Journal of Human-Computer Studies: 2017-2018
- Interaction Studies: 2017
- ACM Transactions on Applied Perception: 2017

CONFERENCE REVIEWING

- RoboCup Symposium: 2018-2026. Session Chair in 2022.
- Cognitive Science Society Annual Meeting (CogSci): 2026
- ACM/IEEE International Conference on Human-Robot Interaction (HRI) Pioneers Workshop (Doctoral Consortium): 2025
- IEEE Conference on Advanced Robotics and its Social Impact: 2023
- IEEE International Conference on Ubiquitous Robots: 2021
- ACM Conference on Human Factors in Computing Systems (CHI): 2016

- ACM/IEEE International Conference on Human-Robot Interaction (HRI): 2010-2016, 2018, 2021, 2023
- IEEE International Conference on Robotics and Automation (ICRA): 2022-2026
- IEEE International Conference on Robot and Human Interactive Communication (RO-MAN): 2013, 2016, 2017, 2019, 2022, 2024
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2009, 2012, 2017 - 2024
- ASME International Design Engineering Technical Conference & Computers and Information Engineering Conference (IDETC): 2019
- Conference on Robot Learning (CoRL): 2018-2019
- Robotics, Science, and Systems (RSS): 2016
- ICDL-Epirob (formerly ICDL and Epirob): 2012, 2013, 2015-2016
- International Conference on Development and Learning (ICDL): 2008-2010
- IEEE-RAS International Conference on Humanoid Robots (HUMANOIDS): 2009-2010
- International Conference on Epigenetic Robotics (EPIROB): 2009
- IEEE International Conference on Automatic Face and Gesture Recognition: 2014
- Association for the Advancement of Artificial Intelligence Undergraduate Consortium (AAAI-UC): 2024, 2025
- The European Conference on Artificial Intelligence (ECAI): 2006, 2024

OTHER ACTIVITIES

- Faculty Advisor - ECLAIR (Engineering and Computational Learning of Artificial Intelligence in Robotics), UT Austin Computer Science Robotics Club: 2023-present.
- Reviewer - Texas Robotics Minor Applications - 2026
- Bridging Barriers - UT Good Systems - Living and Working with Robots - Co-Director (Alongside Elliott Hauser): 2021-present
- UT Austin Ethical Artificial Intelligence Program (and graduate portfolio program) - Convergent, Responsible, and Ethical AI Training Experience for Roboticists (CREATE Roboticists) Co-PI: 2021-present
- NSF Grant Review Panelist: 2017-2019
- Texas Robotics Portfolio Seminar Organizer: 2019-Spring 2024 (Gave organization responsibilities to Michelle Garel in 2022, ran the seminar session until Spring 2024.)
- UT Austin Undergraduate Research Forum Judge: 2018-present
- UT Austin Villa @ Home RoboCup@Home Team Co-Supervisor: 2017-2019, 2022

- Texas Robotics Academy Instructor: 2018-2025
- Freshman Research Initiative Summer Fellowship Reviewer: 2018-2021
- Computer Science Department Graduation Representative: 2021

Publications

JOURNAL ARTICLES

1. Yao-Cheng Chan, Elliott Hauser, Sadanand Modak, Joydeep Biswas, and Justin Hart. Perceived Social Intelligence and Human Compliance in Incidental Human-Robot Encounters. *Expert Systems*. vol. 43(8), article e70332. July 6, 2026. <https://doi.org/10.1111/exsy.70332>
2. Anthony Francis, Claudia Pérez-D'Arpino, Chengshu Li, Fei Xia, Alexandre Alahi, Rachid Alami, Aniket Bera, Abhijat Biswas, Joydeep Biswas, Rohan Chandra, Hao-Tien Chiang, Michael Everett, Sehoon Ha, Justin Hart, Jonathan How, Haresh Karnan, Tsang-Wei Lee, Luis Manso, Reuth Mirsky, Sören Pirk, Phani Teja Singamaneni, Peter Stone, Ada Taylor, Peter Trautman, Nathan Tsoi, Marynel Vázquez, Xuesu Xiao, Peng Xu, Naoki Yokoyama, Alexander Toshev, Roberto Martín-Martín. Principles and Guidelines for Evaluating Social Robot Navigation Algorithms. *ACM Transactions on Human-Robot Interaction (THRI)*. vol. 14(2), article 34, 65 pages. February 20, 2025. <https://doi.org/10.1145/3700599>
3. Reuth Mirsky, Xuesu Xiao, Justin Hart, and Peter Stone. Conflict Avoidance in Social Navigation—a Survey. *ACM Transactions on Human-Robot Interaction (THRI)*. vol. 13(1), article 13, 36 pages. March 11, 2024. <https://doi.org/10.1145/3647983>
4. Alessandra Rossi, Maike Paetzel-Prüsmann, Merel Keijzers, Michael Anderson, Susan Leigh Anderson, Daniel Barry, Jan Gutsche, Justin Hart, Luca Iocchi, Ainse Kokkelmans, Wouter Kuijpers, Yun Liu, Daniel Polani, Caleb Roscon, Marcus Scheunemann, Peter Stone, Florian Vahl, René van de Molengraft, and Oskar von Stryk. The human in the loop: Perspectives and challenges for RoboCup 2050. *Autonomous Robots*. May 16, 2024. vol. 48, article 8. doi:10.1007/s10514-024-10159-3
5. Haresh Karnan, Anirudh Nair, Xuesu Xiao, Garrett Warnell, Soren Pirk, Alexander Toshev, Justin Hart, Joydeep Biswas, and Peter Stone. Socially CompliAnt Navigation Dataset (SCAND): A Large-Scale Dataset Of Demonstrations For Social Navigation. In *IEEE Robotics and Automation Letters*. vol 7(4). pp. 11807 - 11814 (Presented at IROS 2022). October 2022. doi:10.1109/LRA.2022.3184025
6. Jesse Thomason, Aishwarya Padmakumar, Jivko Sinapov, Nick Walker, Yuqian Jiang, Harel Yedidion, Justin Hart, Peter Stone, and Raymond Mooney. Jointly improving parsing and perception for natural language commands through human-robot dialog. In *Journal of Artificial Intelligence Research*. vol. 67. pp. 327-374. February 26, 2020. doi:10.1613/jair.1.11485
7. Wilma A. Bainbridge, Justin W. Hart, Eli S. Kim, and Brian Scassellati. (2010) The Benefits of Interactions with Physically Present Robots Over Video-Displayed Agents. *International Journal of Social Robotics*. vol. 3(1), pp. 41 - 52. Published online October 16, 2010. doi:10.1007/s12369-010-0082-7

CONFERENCE PAPERS

1. Nicholas Verzic, Abhinav Chadaga, Justin Hart. Recovering Missed Detections in an Elevator Button Segmentation Task. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Abu Dhabi, United Arab Emirates. October 14-18, 2024. pp. 13355–13362. doi:10.1109/IROS58592.2024.10801499
2. Elliott Hauser, Yao-Cheng Chan, Sadanand Modak, Joydeep Biswas, Justin Hart. Vid2Real HRI: Align video-based HRI study designs with real-world settings. In *Proceedings of the IEEE RAS International Conference on Robot and Human Interactive Communication (RO-MAN)*. Pasadena, California, USA. August 26-30, 2024. pp. 542–548. doi:10.1109/RO-MAN60168.2024.10731413
3. Carson Stark, Bohkyung Chun, Casey Charleston, Varsha Ravi, Luis Pabon, Surya Sunkari, Tarun Mohan, Peter Stone, Justin Hart. Dobby: A Conversational Service Robot Driven by GPT-4. In *Proceedings of the IEEE RAS International Conference on Robot and Human Interactive Communication (RO-MAN)*. Pasadena, California, USA. August 26-30, 2024. pp. 1362–1369. doi:10.1109/RO-MAN60168.2024.10731375
4. Elliott Hauser, Yao-Cheng Chan, Ruchi Bhalani, Alekhya Kuchimanchi, Hanaa Siddiqui, Justin Hart. Influencing Incidental Human-Robot Encounters: Expressive movement improves pedestrians’ impressions of a quadruped service robot. In *Proceedings of the Hawaii International Conference on System Sciences (HICSS)*. Waikiki, Hawaii, USA. January 3-6, 2024. pp. 724–733. doi:10.24251/HICSS.2024.087
5. Emily Norman, Keri K. Stephens, Samantha Shorey, Joydeep Biswas, Justin Hart, Elliott Hauser, Luis Sentis. Bringing Training, Learning, and Development into Team Science: Effectively Freeing Team Members from Disciplinary Barriers in Research. In *2023 National Communication Association 109th Annual Convention “Freedom”*. National Harbor, Maryland, USA. November 17, 2023. (Awarded as “Top Papers”)
6. Haresh Karnan*, Anirudh Nair, Xuesu Xiao, Garrett Warnell, Soren Pirk, Alexander Toshev, Justin Hart, Joydeep Biswas, Peter Stone. Socially CompliAnt Navigation Dataset (SCAND): A Large-Scale Dataset Of Demonstrations For Social Navigation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Kyoto, Japan. October 23-27, 2022. (RA-L paper presented at IROS) doi:10.1109/LRA.2022.3184025
7. Blake Holman, Abrar Anwar, Akash Singh, Mauricio Tec, Justin Hart, and Peter Stone. Watch Where You’re Going! Gaze and Head Orientation as Predictors for Social Robot Navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*. Xi’an, China. May 30 - June 5, 2021. pp. 3553–3559. doi:10.1109/ICRA48506.2021.9561286
8. Rishi Shah, Yuqian Jiang, Justin Hart, and Peter Stone. Deep R-Learning for Continual Area Sweeping. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Las Vegas, Nevada, USA (Virtual). October 24, 2020–January 24, 2021. pp. 5542–5547. doi:10.1109/IROS45743.2020.9341626
9. Justin Hart, Reuth Mirsky, Xuesu Xiao, Stone Tejeda, Bonny Mahajan, Jamin Goo, Kathryn Baldauf, Sydney Owen, and Peter Stone. Using Human-Inspired Signals to Disambiguate Navigational Intentions. In *Proceedings of the International Conference on Social Robotics (ICSR)*. Golden, Colorado, USA (Virtual). November 14-18, 2020. pp. 320–331. doi:10.1007/978-3-030-62056-1_27
10. Yuqian Jiang, Nick Walker, Justin Hart, and Peter Stone. Open-World Reasoning for Service Robots. In *Proceedings of the International Conference on Automated Planning and Scheduling*

- (ICAPS). Berkeley, California, USA. July 13-15, 2019. pp. 725–733. doi:10.1609/icaps.v29i1.3541
11. Jesse Thomason, Aishwarya Padmakumar, Jivko Sinapov, Nick Walker, Yuqian Jiang, Harel Yedidsion, Justin Hart, Peter Stone, and Raymond J. Mooney. Improving Grounded Natural Language Understanding through Human-Robot Dialog. In *Proceedings of the International Conference on Robotics and Automation (ICRA)*. Montreal, Quebec, Canada. May 20-24, 2019. pp. 6934–6941. doi:10.1109/ICRA.2019.8794287
 12. Harel Yedidsion, Jacqueline Deans, Connor Sheehan, Mahathi Chillara, Justin Hart, Peter Stone, and Raymond J. Mooney. Optimal Use Of Verbal Instructions For Multi-Robot Human Navigation Guidance. In *Proceedings of the Eleventh International Conference on Social Robotics (ICSR)*. Madrid, Spain. November 26-29, 2019. pp. 133–143. doi:10.1007/978-3-030-35888-4_13
 13. Rolando Fernandez, Nathan John, Sean Kirmani, Justin Hart, Jivko Sinapov, and Peter Stone. Passive Demonstrations of Light-Based Robot Signals for Improved Human Interpretability. In *Proceedings of the IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. Nanjing, China. August 27-31, 2018. pp. 234–239. doi:10.1109/ROMAN.2018.8525728
 14. Justin Hart, Rishi Shah, Sean Kirmani, Nick Walker, Kathryn Baldauf, Nathan John, and Peter Stone. PRISM: Pose Registration for Integrated Semantic Mapping. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Madrid, Spain. October 1-5, 2018. pp. 896–902. doi:10.1109/IROS.2018.8593681
 15. Jesse Thomason, Aishwarya Padmakumar, Jivko Sinapov, Justin Hart, Peter Stone, and Raymond J. Mooney. Opportunistic Active Learning for Grounding Natural Language Descriptions. In *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*. Mountain View, California, USA. November 13-15, 2017. PMLR vol. 78, pp. 67–76.
 16. Justin Hart and Brian Scassellati. Mirror Perspective-Taking with a Humanoid Robot. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. Toronto, Canada. July 22-26, 2012. pp. 1990–1996. doi:10.1609/aaai.v26i1.8389
 17. Justin W. Hart and Brian Scassellati. A Robotic Model of the Ecological Self. In *Proceedings of the IEEE-RAS International Conference on Humanoid Robots (HUMANOIDS)*. Bled, Slovenia. October 26-28, 2011. pp. 682–688. doi:10.1109/HUMANOIDS.2011.6100897
 18. Eleanor Avrunin, Justin Hart, Ashley Douglas, and Brian Scassellati. Effects Related to Synchrony and Repertoire in Perceptions of Robot Dance. In *Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. Lausanne, Switzerland. March 6-9, 2011. pp. 93–100. doi:10.1145/1957656.1957678
 19. Elaine Short, Justin Hart, Michelle Vu, and Brian Scassellati. No Fair!! An Interaction with a Cheating Robot. In *Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. Osaka, Japan. March 2-5, 2010. **(Nominated for best paper award)**. pp. 219–226. doi:10.1109/HRI.2010.5453193
 20. Wilma A. Bainbridge, Justin Hart, Eli S. Kim, and Brian Scassellati. The Effect of Presence on Human-Robot Interaction. In *Proceedings of the IEEE International Conference on Robot and Human Interactive Communication (ROMAN)*. Munich, Germany. August 1-3, 2008. pp. 701–706. doi:10.1109/ROMAN.2008.4600749

21. Ashish Sabharwal, Carlos Ansotegui, Carla P. Gomes, Justin Hart, and Bart Selman. QBF Modeling: Exploiting Player Symmetry for Simplicity and Efficiency. In *Proceedings of the International Conference on Theory and Applications of Satisfiability Testing (SAT)*. Seattle, Washington, USA, August 12-15, 2006. pp. 382–395. doi:10.1007/11814948_35

WORKSHOP PAPERS

1. Ryan Lee, Justin Hart, Robert Blake Anderson, Corrie Van Sice, Andrew Sharp, and Mitch Pryor. Online Robot Parameter Tuning from Demonstration and Multi-Modal Context Classification. In *Proceedings of the First Workshop on Semantic-Aware Mapping and Navigation: Towards Cognitive and Adaptive Robots (SeMaNa) at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Pittsburgh, Pennsylvania, USA. September 27, 2026.
2. Elliott Hauser, Justin Hart, Samuel Baker, Luis Sentis. Embedding spatiotemporal context for conversations with autonomous mobile robots. In *Proceedings of the HRI Workshop on Human-Robot Conversational Interaction*. Stockholm, Sweden. March 13, 2023.
3. Haresh Karnan, Anirudh Nair, Xuesu Xiao, Garrett Warnell, Soren Pirk, Alexander Toshev, Justin Hart, Joydeep Biswas, and Peter Stone. Socially CompliAnt Navigation Dataset (SCAND): A Large-Scale Dataset Of Demonstrations For Social Navigation. (Extended Abstract) In *Proceedings of the ICRA Workshop on Social Robot Navigation: Advances and Evaluation*. Philadelphia, Pennsylvania, USA. May 27, 2022.
4. Parth Chonkar, Geethika Hemkumar, Huihai Wang, Daksh Dua, Shikhar Gupta, Yao-Cheng Chan, Justin Hart, Elliott Hauser, Reuth Mirsky, Joydeep Biswas, Junfeng Jiao, and Peter Stone. Look to my Lead: How Does a Leash Affect Perceptions of a Quadruped Robot?. In *Proceedings of the ICRA Workshop on Social Robot Navigation: Advances and Evaluation*. Philadelphia, Pennsylvania, USA. May 27, 2022.
5. Justin Hart, Reuth Mirsky, Xuesu Xiao, and Peter Stone. Incorporating Gaze into Social Navigation. In *Proceedings of the Robotics: Science and Systems Workshop on Social Robot Navigation*. Virtual, July 13, 2021.
6. Justin Hart, Sara Sheikholeslami, Matthew K. X. J. Pan, Wesley P. Chan, and Elizabeth A. Croft. Predictions of Human Task Performance and Handover Trajectories for Human-Robot Interaction. In *Proceedings of the HRI 2015 Workshop on Human-Robot Teaming*. Portland, Oregon, USA. March 2, 2015.
7. Justin Hart, Brian Gleeson, Matthew K. X. J. Pan, AJung Moon, Karon MacLean, and Elizabeth A. Croft. Gesture, Gaze, Touch, and Hesitation: Timing Cues for Collaborative Work. In *Proceedings of the HRI Workshop on Timing in Human-Robot Interaction 2014*. Bielefeld, Germany. March 3, 2014.
8. Justin Hart and Brian Scassellati. Robotic Self-Models Inspired by Human Development. In *Proceedings of the AAAI-10 Workshop on Metacognition for Robust Social Systems*. Atlanta, Georgia, USA. July 11, 2010.
9. Justin Hart, Brian Scassellati, and Steven W. Zucker. Epipolar Geometry for Humanoid Robotic Heads. In *Proceedings of the 4th International Cognitive Vision Workshop*. Santorini, Greece. May 12, 2008.

SYMPOSIUM PAPERS

1. Samuel Baker, Elliott Hauser and Justin Hart. Trustworthy Memory Practices for Better Living and Working with Robots. In *Proceedings of the First International Symposium on Trustworthy Autonomous Systems*. Edinburgh, UK, July 11-12, 2023. pp. 1–5. doi:10.1145/3597512.3597516
2. Elliott Hauser, Yao-Cheng Chan, Geethika Hemkumar, Daksh Dua, Parth Chonkar, Efren Mendoza Enriquez, Tiffany Kao, Shikhar Gupta, Huihai Wang, Justin Hart, Reuth Mirsky, Joydeep Biswas, Junfeng Jiao and Peter Stone. “What’s That Robot Doing Here?”: Perceptions Of Incidental Encounters With Autonomous Quadruped Robots. In *Proceedings of the First International Symposium on Trustworthy Autonomous Systems*. Edinburgh, UK, July 11-12, 2023. pp. 1–15. doi:10.1145/3597512.3599707
3. Anthony Francis, Claudia Perez-D’Arpino, Chengshu Li, Fei Xia, Alexandre Alahi, Aniket Bera, Abhijat Biswas, Joydeep Biswas, Hao-Tien Lewis Chiang, Michael Everett, Sehoon Ha, Justin Hart, Haresh Karnan, Tsang-Wei Edward Lee, Luis Manso, Reuth Mirsky, Soren Pirk, Phani Teja Singamaneni, Peter Stone, Ada Taylor, Peter Trautman, Nathan Tsoi, Marynel Vazquez, Xuesu Xiao, Peng Xu, Naoki Yokoyama, Roberto Martin-Martin, Alexander Toshev. Benchmarking Social Robot Navigation Across Academia and Industry. In *Proceedings of the AAAI Spring Symposium on HRI in Academia and Industry: Bridging the Gap*. AAAI Spring Symposium Series. Palo Alto, California, USA, March 27-29, 2023.
4. Yao-Cheng Chan, Justin Hart, and Elliott Hauser. Understanding Human-Robot Encounters: Challenges for Industry and Science. In *Proceedings of the AAAI Spring Symposium on HRI in Academia and Industry: Bridging the Gap*. AAAI Spring Symposium Series. Palo Alto, California, USA, March 27-29, 2023.
5. Minkyu Kim, Miguel Arduengo, Nick Walker, Yuqian Jiang, Justin Hart, Peter Stone, and Luis Sentis. Active Target Search for Autonomous Person Following. In *Proceedings of the AAAI Spring Symposium on Machine Learning for Mobile Robot Navigation in the Wild*. AAAI Spring Symposium Series. Palo Alto, California, USA (Virtual), March 22-24, 2021.
6. David Balaban, Harshavardhan Jagannathan, Henry Liu, and Justin Hart, Efficient Placard Discovery for Semantic Mapping During Frontier Exploration. In *Proceedings of the AAAI Spring Symposium on Machine Learning for Mobile Robot Navigation in the Wild*. AAAI Spring Symposium Series. Palo Alto, California, USA (Virtual), March 22-24, 2021.
7. Asha Kailin Jain, Maxwell Svetlik, Nicholas Machak, Kavan Singh Sikand, Cem Karamanli, Kaiyu Zhou, Justin Hart, Joydeep Biswas, Luis Sentis, Junfeng Jiao. An Open-Source Framework for Last Mile Delivery with Heterogeneous Robots. In *Proceedings of the AAAI Spring Symposium on Machine Learning for Mobile Robot Navigation in the Wild*. AAAI Spring Symposium Series. Palo Alto, California, USA (Virtual). March 22-24, 2021.
8. Rishi Shah, Yuqian Jiang, Haresh Karnan, Gilberto Briscoe-Martinez, Dominick Mulder, Ryan Gupta, Rachel Schlossman, Marika Murphy, Justin Hart, Luis Sentis, Peter Stone. Solving Service Robot Tasks: UT Austin Villa @ Home 2019 Team Report. In *Proceedings of Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI)*. AAAI Fall Symposium Series. Arlington, Virginia, USA. November 7-9, 2019.
9. Justin Hart, Reuth Mirsky, Stone Tejada, Bonny Mahajan, Jamin Goo, Kathryn Baldauf, Sydney Owen, Peter Stone. Unclogging Our Arteries: Using Human-Inspired Signals to Disambiguate Navigational Intentions. In *Proceedings of Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI)*. AAAI Fall Symposium Series. Arlington, Virginia, USA. November 7-9, 2019.

10. Yuqian Jiang, Nick Walker, Minkyu Kim, Nicolas Brissonneau, Daniel S. Brown, Justin Hart, Scott Niekum, Luis Sentis, and Peter Stone. LAAIR: A Layered Architecture for Autonomous Interactive Robots. In *Proceedings of the Reasoning and Learning in Real-World Systems for Long-Term Autonomy (LTA) symposium*, AAAI Fall Symposium Series. Arlington, Virginia, USA. October 18-20, 2018.
11. Justin Hart, Harel Yedidsion, Yuqian Jiang, Nick Walker, Rishi Shah, Jesse Thomason, Aishwarya Padmakumar, Rolando Fernandez, Jivko Sinapov, Raymond Mooney, Peter Stone. Interaction and Autonomy in RoboCup@Home and Building-Wide Intelligence. In *Proceedings of the Artificial Intelligence (AI) for Human-Robot Interaction (HRI) symposium (AI-HRI)*. AAAI Fall Symposium Series. Arlington, Virginia, USA. October 18-20, 2018.

PRE-PRINTS

1. Elliott Hauser, Yao-Cheng Chan, Sadanand Modak, Joydeep Biswas, Justin Hart. Vid2Real HRI: Align video-based HRI study designs with real-world settings. In *arXiv preprint arXiv:2403.15798*. March 23, 2024.
2. Elliott Hauser, Yao-Cheng Chan, Ruchi Bhalani, Alekhya Kuchimanchi, Hanaa Siddiqui, Justin Hart. Influencing Incidental Human-Robot Encounters: Expressive movement improves pedestrians' impressions of a quadruped service robot. In *arXiv preprint arXiv:2311.04454*. November 8, 2023.
3. Carson Stark, Bohkyung Chun, Casey Charleston, Varsha Ravi, Luis Pabon, Surya Sunkari, Tarun Mohan, Peter Stone, Justin Hart. Dobby: A Conversational Service Robot Driven by GPT-4. In *arXiv preprint arXiv:2310.06303*. October 10, 2023.
4. David Balaban, Justin Medich, Pranay Gosar, Justin Hart. Propagating Semantic Labels in Video Data. In *arXiv preprint arXiv:2310.00783*. October 1, 2023.
5. Hugo Latapie, Shan Yu, Patrick Hammer, Kristinn R. Thorisson, Vahagn Petrosyan, Brandon Kynoch, Alind Khare, Payman Behnam, Alexey Tumanov, Aksheit Saxena, Anish Aralikatti, Han-ning Chen, Mohsen Imani, Mike Archbold, Tangrui Li, Pei Wang, Justin Hart. Ethosight: A Reasoning-Guided Iterative Learning System for Nuanced Perception based on Joint-Embedding & Contextual Label Affinity. In *arXiv preprint arXiv:2307.10577*. July 20, 2023.
6. Anthony Francis, Claudia Perez-D'Arpino, Chengshu Li, Fei Xia, Alexandre Alahi, Rachid Alami, Aniket Bera, Abhijat Biswas, Joydeep Biswas, Rohan Chandra, Hao-Tien Lewis Chiang, Michael Everett, Sehoon Ha, Justin Hart, Jonathan P How, Haresh Karnan, Tsang-Wei Edward Lee, Luis J Manso, Reuth Mirsky, Soeren Pirk, Phani Teja Singamaneni, Peter Stone, Ada V Taylor, Peter Trautman, Nathan Tsoi, Marynel Vazquez, Xuesu Xiao, Peng Xu, Naoki Yokoyama, Alexander Toshev, Roberto Martin-Martin. Principles and Guidelines for Evaluating Social Robot Navigation Algorithms. *arXiv preprint arXiv:2306.16740*. June 29, 2023.
7. David Balaban, Justin Hart. Automatic Sign Reading and Localization for Semantic Mapping with an Office Robot. In *arXiv preprint arXiv:2209.11432*. September 23, 2022.
8. Reuth Mirsky, Xuesu Xiao, Justin Hart, Peter Stone. Conflict Avoidance in Social Navigation – A Survey. *arXiv preprint arXiv:2106.12113*. June 23, 2021.

BOOK CHAPTERS

1. Justin Hart, Sara Sheikholeslami, Brian Gleeson, Elizabeth Croft, Karon MacLean, Frank P. Ferrie, Clément Gosselin and Denis Laurandean (2018) Developing Robot Assistants with Communicative Cues for Safe, Fluent HRI. In Hussein A. Abbass, Jason Scholz, and Darryn J. Reid (Eds.). *Foundations of Trusted Autonomy*. Cham, Switzerland: Springer. pp. 247–270. doi:10.1007/978-3-319-64816-3_14
2. Justin Hart and Brian Scassellati. (2014) Robotic Self-Modeling. In Jeremy Pitt (Ed.). *The Computer After Me*. London, UK: Imperial College Press. pp. 207–218. doi:10.1142/9781783264186_0014
3. Justin Hart and Brian Scassellati. (2011) Robotic Models of Self. In Michael T. Cox, and Anita Raja (Eds.). *Metareasoning: Thinking about Thinking*. Cambridge, Massachusetts, USA: MIT Press. pp. 283–294. doi:10.7551/mitpress/9780262014809.003.0018

NEWSLETTER ARTICLES

1. Justin Hart and Brian Scassellati. Self-Awareness and Social Competencies. *AMD NEWSLETTER: The Newsletter of the Autonomous Mental Development Technical Committee*. vol. 12(1), Spring 2015. pp. 7–8.

DOCTORAL THESIS

1. Justin Hart. (2014) Robot Self-Modeling. Ph.D. Thesis, Yale University, New Haven, Connecticut, USA, December 2014.

Talks and Posters

TALKS TO CORPORATE LABORATORIES

1. Justin Hart. *Semantic SLAM for Augmented Reality and Robotics. (Final Readout)* Cisco Research. September 9, 2024.
2. Justin Hart and Yifan Xu. *Dobby: An Interactive Robot Incorporating Large Language Models* UT Austin Good Systems Symposium. March 27, 2024. <https://www.youtube.com/watch?v=hnrxhxFRfk8>
3. Carson Stark & Justin Hart. *Dobby at Hook 'Em House at Antones - UT at SXSW* SXSW. March 8, 2024.
4. Justin Hart. *Semantic SLAM for Augmented Reality and Robotics. (Contract 1-Year End Talk)* Cisco Research. October 16, 2023
5. Justin Hart. *Semantic SLAM for Augmented Reality and Robotics. (Update 2)* Cisco Research. June 12, 2023
6. Justin Hart. *Semantic SLAM for Augmented Reality and Robotics. (Update 1)* Cisco Research. December 1, 2022
7. Justin Hart. *Vision, Mapping, and Interaction for Home Robots & Human Interfaces*. Cisco Research. October 13, 2022

TALKS AT UT AUSTIN

1. Justin Hart. *Moving Robots from Factories and Warehouses and Into Homes and Offices*. Texas Symposium on Machine Learning, Responsible AI & Robotics. March 4, 2026.
2. Justin Hart. *Moving Robots from Factories and Warehouses and into Homes and Offices*. Moonshot Ideas for AI & Robotics in Dementia Prevention & Care. February 27, 2026.
3. Justin Hart (Panelist). *Moonshot Ideas Panel Discussion*. Moonshot Ideas for AI & Robotics in Dementia Prevention & Care. February 27, 2026.
4. Justin Hart (Panel Moderator). *Responsible AI and Interactive Robots*. Responsible AI in 2025. Panelists: Andreea Bobu (MIT), Reuth Mirsky (Tufts), and Nathan Tsoi (UT Austin). November 18, 2025.
5. Justin Hart. *The Living with Robots Lab*. CS 398T. October 23, 2025.
6. Justin Hart. Robotics FRI Streams Event: *The Living with Robots Lab*. October 15, 2025.
7. Justin Hart. *The Living with Robots Lab* Dean's Scholars Honors Program at The University of Texas at Austin - Dean's Scholars Lunch Talk. October 11, 2024
8. Justin Hart. *The Living with Robots Lab* Texas Robotics Symposium. October 19, 2023
9. Justin Hart. *The Living with Robots Lab* Engineering and Computational Learning of Artificial Intelligence in Robotics (ECLAIR). October 5, 2023
10. Justin Hart. *Living and Working with Robots* Good Systems Fall Kickoff Meeting. September 9, 2023

INVITED TALKS⁴

1. Justin Hart. *Moving Robots from Factories and Warehouses into Homes and Offices*. 2026 Training & Development Expo, Austin Development Services, City of Austin. Austin, Texas, USA. April 22, 2026.
2. Justin Hart. *Moving Robots from Factories and Warehouses and Into Homes and Offices*. Contextual Robotics Institute, University of California San Diego. La Jolla, California, USA. February 19, 2026.
3. Justin Hart. *Moving Robots from Places Designed for Robots to Places Designed for People*. Guest lecture, INFO 5356-030: Introduction to Human-Robot Interaction, Cornell Tech, Cornell University. New York, New York, USA (Virtual). November 12, 2025.
4. Justin Hart. Domestic Robots in The Living with Robots Laboratory. *Austin Robotics Meetup*. Online. March 5, 2025.
5. Justin Hart. The Living with Robots Laboratory. *National Science Foundation Quori Community Workshop*. Madison, Wisconsin, USA. April 4, 2025.
6. Justin Hart. (Panelist) Design and Collection of HRI Datasets. *HRI Workshop on The Road to Reliable Robots: Interpretable, Accessible, and Reproducible HRI Research*. Melbourne, Victoria, Australia. March 3, 2025.

⁴Not including department talks or paper presentations

7. Justin Hart. Living and Working with Reliable Robots. *HRI Workshop on The Road to Reliable Robots: Interpretable, Accessible, and Reproducible HRI Research*. Melbourne, Victoria, Australia. March 3, 2025.
8. Justin Hart (Panelist). *HRI Workshop on End User Development for Human-Robot Interaction (EUD4HRI)*. Boulder, Colorado, USA. March 15, 2024.
9. Justin Hart (Panelist). *AAAI Fall Symposium on Distributed Teaching Collaboratives for AI and Robotics*. Arlington, Virginia, USA. November 17, 2022.
10. Justin Hart. *Getting Robots into Every Home and Workplace. CAIDA, University of British Columbia*. Online. June 21, 2022.
11. Justin Hart. Social Cues for Social Navigation *Google's HRI for Learning Robots Workshop*. Online. June 1-2, 2022.
12. Justin Hart. What's Going on at the UT Lab? *Austin Robotics Meetup*. Online. May 19, 2022.
13. Justin Hart. Robotics, Artificial Intelligence, and Human-Robot Interaction - Guest lecture on behalf of TIDES/FRI. *King Abdullah University of Science and Technology (KAUST)*. Online. June 27, 2021.
14. Justin Hart. What's Going on at the UT Austin Robotics Lab? *Austin Robotics Meetup*. Online. October 10, 2020.
15. Junfeng Jiao, Justin Hart, Joydeep Biswas, and Luis Sentis. (Moderated by Peter Stone). Robotics Panel: Last Mile Autonomous Delivery Systems. *The University of Texas at Austin - Bridging Barriers - Good Systems*. Online. September 11, 2020.
16. Justin Hart. Signaling to Peers Using Human-Inspired Communicative Cues. *Bithacks*. Online. August 8, 2020.
17. Justin Hart. Signaling to Peers Using Human-Inspired Communicative Cues. *Virtual Humanoid RoboCup Open Workshop*. Online. June 6, 2020.
18. Justin Hart. Semantic Mapping and Service Robots. *IEEE Metrocon*. Hurst, Texas, USA. November 7, 2018.
19. Justin Hart. Robot Self-Modeling and Self-Other Reasoning. *Mensa National Gathering*. Vancouver, British Columbia, Canada. August 1, 2015.
20. Justin Hart. Robot Self-Modeling and Self-Other Reasoning. *UBC Postdoc Talks*. Vancouver, British Columbia, Canada. July 8, 2015.
21. Justin Hart. Learning about people to build better robots. *Creative Mornings: Vancouver*. Vancouver, BC, Canada, May 1, 2015.
22. Justin Hart. Robotic Self-Modeling. *Ideacity*. Toronto, Ontario, Canada. June 2013.
23. Justin Hart. Robotic Self-Modeling. *Society of Manufacturing Engineers Annual Conference*. Baltimore, Maryland, USA. June 2013.
24. Justin Hart. Robot Self-Modeling. *UT Austin Forum for Artificial Intelligence*. Austin, Texas, USA. April 19, 2013.

CONFERENCE AND WORKSHOP TALKS (INCLUDING INVITED TALKS)

1. Justin Hart. *Moving Robots from Factories and Warehouses and Into Homes and Offices*. Quori Community Meeting. (Virtual). February 17, 2026.
2. Justin Hart. Ubiquitous HRI and Social Navigation. *AAAI Fall Symposium on Artificial Intelligence for Human-Robot Interaction (AI-HRI)*. (Invited Talk). Online. November 4, 2021.
3. Justin Hart. Robot Self-Modeling. *The Development of the Self: from self-perception to interaction under uncertainty. Workshop at ICDL-Epirob*. (Invited Talk). Lisbon Portugal, September 18, 2017.
4. Justin Hart and Brian Scassellati. Creating Social Agency. *NSF-JST workshop on Human-Robot Interaction*. (Invited Talk). Menlo Park, California, USA, August 2010.
5. Justin Hart, Brian Scassellati, and Steven Zucker. Estimating the Kinematics of Unseen Joints that Affect the Stereo Vision System. *The 4th Annual New England Manipulation Symposium*. Providence, Rhode Island, USA, May 2008.

POSTERS

1. Justin Hart, Brian Scassellati, and Steven W. Zucker. Calibrating the Eye Motion of a Humanoid Robot. Appeared at *The IEEE International Conference on Development and Learning (ICDL)*. Monterey, California, USA. August 9-12, 2008.
2. Justin Hart, Eleanor Avrunin, David Golub, Brian Scassellati, and Steven Zucker. Incorporating Active Vision into the Body Schema. Appeared at *The ACM/IEEE International Conference on Human-Robot Interaction (HRI 2009)*. La Jolla, California, USA, March 11-13, 2009. pp. 315–316. doi:10.1145/1514095.1514193
3. Justin Hart, Brian Scassellati, and Steven Zucker. Epipolar Geometry for Humanoid Robotic Heads. Appeared at *The Third North East Student Colloquium on Artificial Intelligence (NESCAI)*. Ithaca, New York, USA, May 2008.

UNDERGRADUATE RESEARCH FORUM POSTERS

1. Praneel Seth, Kunal Tiwari, Mukund Raman, Akshay Gaitonde, Arshdeep Gupta, Brandon Nunley, David Balaban, and Justin Hart. Predicting human motion for socially-aware robot navigation. Poster, *Tech & Science Undergraduate Research Forum (URF)*, College of Natural Sciences, The University of Texas at Austin. Austin, Texas, USA. April 17, 2026.
2. Lucas Helms, Wyatt Hansen, Mukund Maini, Matthew Builes, Bhavana Katta, and Justin Hart. Diffusion-based motion prediction for natural human-robot handovers. Poster, *Tech & Science Undergraduate Research Forum (URF)*, College of Natural Sciences, The University of Texas at Austin. Austin, Texas, USA. April 17, 2026.
3. Carson Stark, Tanay Garg, Ayon Das, Harshal Bharatia, Sherwin Amal, and Justin Hart. Agent-in-the-Loop Execution for Conversational Service Robots. Poster, *Tech & Science Undergraduate Research Forum (URF)*, College of Natural Sciences, The University of Texas at Austin. Austin, Texas, USA. April 17, 2026.

UNDERGRADUATE RESEARCH REPORTS

1. Carson Stark, Tanay Garg, Ayon Das, Harshal Bharatia, Sherwin Amal, and Justin Hart. *Agent in the Loop: Integrating Code-as-Policy with Runtime LLM Control for Service Robots*. CS 370 research report, Department of Computer Science, University of Texas at Austin. Austin, Texas, USA. May 2026. 10 pp.
2. Jacob Mathew, Brandon Nunley, Ananya Chintalapudi, Akshay Gaitonde, Mukund Raman, Praneel Seth, Kunal Tiwari, Justin Hart, and Nick Fey. *“FUTURE”: A Foundational Unified Transformer for Unconstrained, Real-time Extrapolation of Human Motion*. CS 370 research report, Department of Computer Science, University of Texas at Austin. Austin, Texas, USA. May 2025. 5 pp.

Selected Media Coverage

1. *NSF STC: Center for Human and Robot Co-Adaptation*
 - (a) UT News. *UT To Lead NSF Center That Will Study How Robots and People Learn To Live and Work Together*. August 27, 2026.
<https://news.utexas.edu/2026/08/27/ut-to-lead-nsf-center-that-will-study-how-robots-and-people-learn-to-live-and-work-together/>
 - (b) National Science Foundation. *NSF launches three new Science and Technology Centers with \$90M investment in transformative research*. August 26, 2026.
<https://www.nsf.gov/news/nsf-launches-three-new-science-technology-centers-90m>
 - (c) Joydeep Biswas (LinkedIn). *Center for Human and Robot Co-Adaptation launch announcement*. 2026.
https://www.linkedin.com/posts/joydeep-biswas_ut-to-lead-nsf-center-activity-7499088454455181312-RTKP
2. *Texas Robotics Academy*
 - (a) Texas Robotics. *High School Students Dive Into Robotics at UT Austin’s Academy for Robotics*. July 1, 2025.
<https://robotics.utexas.edu/news/318>
3. *The Living with Robots Laboratory and RoboCup*
 - (a) Austin American-Statesman. *Parade of robots takes over UT Austin campus to mark end of ‘Year of AI’*. November 18, 2024.
<https://www.statesman.com/story/business/technology/2024/11/18/ut-austin-year-of-ai-robot-parade-texas-university/76306951007/>
 - (b) UT Austin Computer Science (coverage of KXAN reporting). *Could a Robot Win the World Cup? UT Experts Explore Future of Automations*. April 19, 2024.
<https://www.cs.utexas.edu/news/2024/could-robot-win-world-cup-ut-experts-explore-future-automations>
4. *NSF GCR: Community-Embedded Robotics: Understanding Sociotechnical Interactions with Long-term Autonomous Deployments*

- (a) CBS News. *Texas program studies human-robot interactions*. February 9, 2023.
<https://www.cbsnews.com/video/texas-program-studies-human-robot-interactions/>
 - (b) CBS Austin. *Robots will soon be among students at the University of Texas*. October 18, 2022.
<https://cbsaustin.com/news/local/robots-will-soon-be-among-students-at-the-university-of-texas>
 - (c) Texas Monthly. *It's Okay to Fear the Robot Dogs Coming to UT*. October 27, 2022.
<https://www.texasmonthly.com/news-politics/robot-dogs-coming-to-ut/>
 - (d) USA Today/Austin American-Statesman. *Dog-like robots will make deliveries on a Texas campus to see how people, robots interact*. October 18, 2022.
<https://www.usatoday.com/story/news/education/2022/10/18/university-texas-dog-robots-deliver/10532562002/>
 - (e) KVUE (ABC Affiliate). *New dog-like robots to appear on UT campus*. October 17, 2022.
<https://www.youtube.com/watch?v=30ITL5i9PnM>
 - (f) UT News. *Can Robots and Humans Co-exist in Public? UT Campus Study Will Offer Answers*. October 17, 2022.
<https://news.utexas.edu/2022/10/17/can-robots-and-humans-co-exist-in-public-ut-campus-study-will-offer-answers/>
5. *NSF NRT: Convergent, Responsible, and Ethical AI Training Experience in Robotics*
- (a) UT College of Natural Sciences. *New Partnership Will Scale Up Investment in Ethical AI Research and Innovation*. August 5, 2022.
<https://cns.utexas.edu/news/new-partnership-will-scale-up-investment-in-ethical-ai-research-and-innovation>
6. *Interview on Robotics Research*
- (a) Grit Ventures / Medium. *Interview with Grit Ventures Academic Council Member, Dr. Justin Hart, Assistant Professor of Practice, UT Austin*. April 11, 2021.
<https://grit-ventures.medium.com/interview-with-academic-council-member-dr-8cd2aa403e72>
7. *The Short-to-Medium Range Autonomous Delivery System*
- (a) Medium. *Robots in Real Time*. September 21, 2020.
<https://medium.com/good-systems/robots-in-real-time-c914d1fe2fe2>
8. *The Opening of the Anna Hiss Gymnasium and Establishment of the Army Futures Command Robotics Center of Excellence (AFC-RCOE)*
- (a) UT News. *Mission Accomplished*. November 5, 2020.
<https://news.utexas.edu/2020/11/05/mission-accomplished/>
9. *My doctoral thesis - A robot performing motor skills in a mirror*
- (a) NBC News. *What the Rise of Sentient Robots Will Mean for Human Beings*. June 19, 2017.
<https://www.nbcnews.com/mach/tech/what-rise-sentient-robots-will-mean-human-beings-ncna773146>

- (b) New Scientist. *Robot learns to recognise itself in the mirror*. August 22, 2012.
<http://www.newscientist.com/article/mg21528785.900-robot-learns-to-recognise-itself-in-the-mirror.html>
- (c) BBC News. *Robot learns to recognise itself in mirror*. August 23, 2012.
<http://www.bbc.com/news/technology-19354994>
- (d) NBC News. *'Robot learns to track itself and the world through a mirror*. August 25, 2012.
<http://www.nbcnews.com/technology/futureoftech/robot-learns-track-itself-world-through-mirror-961379>
- (e) Business Standard. *Now, a 'Self-aware' robot that recognises itself in the mirror*. August 23, 2012.
<http://tinyurl.com/9txnky7>
- (f) Technabob. *Nico the Robot Recognizes Self in the Mirror*. August 24, 2012.
<https://technabob.com/blog/2012/08/24/nico-the-robot-recognizes-self/>
- (g) CBS SmartPlanet. *Robot passes one milestone in tests of self-awareness*. August 31, 2012.
<http://www.smartplanet.com/blog/science-scope/robot-passes-one-milestone-in-tests-of-self-awareness/13600>
- (h) El Mundo. *'Nico, el robot que está aprendiendo a mirarse al espejo*. August 29, 2012.
<http://www.elmundo.es/elmundo/2012/08/29/navegante/1346229933.html>
- (i) Yale Graduate School of Arts & Sciences: Graduate School News and Events, "Using Robots to Study Self Awareness," April, 2012.
<http://www.yale.edu/graduateschool/publications/news/201205/computer-science-robots-self-awareness.html>

10. *No Fair! An Interaction with a Cheating Robot*

- (a) GE Focus Forward Films. *Robot*. June, 2012.
<http://focusforwardfilms.com/films/41/robot> (Also featured in Google Solve for X.)

Awards & Recognitions

- Society of Manufacturing Engineers Innovation Watch List - 2013

Graduate Mentoring

- PhD students:
 - David Balaban. Thesis Project: Semantic Mapping
Supported by Texas Robotics IAP funding awarded to Justin Hart and Nick Fey.
- Committee Member:
 - Christina Petlowany. Ph.D. dissertation: *Autonomous robots with gaze-based user supervision*. Mechanical Engineering, The University of Texas at Austin. Advisers: Mitch Pryor and Carolyn Seepersad. August 2025.
 - Huihai Wang. Ph.D. dissertation: *Autonomous delivery robots for off-peak-hour urban freight: integration strategies for complex urban environments*. Community and Regional Planning, The University of Texas at Austin. Adviser: Junfeng Jiao. May 2026.

- Brandon Nunley. Ph.D. dissertation: *Multi-modal assessment and optimization of hip preservation surgery: integrating patient data, gait biomechanics, and computational modeling*. Biomedical Engineering, The University of Texas at Austin. Adviser: Nicholas Fey. May 2026.
Supported by Texas Robotics IAP funding awarded to Justin Hart and Nick Fey.
- Zahra Rezaei Khavas. External Ph.D. dissertation committee, University of Massachusetts Lowell. *Trust Beyond Performance: Understanding Moral and Performance Trust Violations in Human–Robot Interaction*. Defense: April 3, 2026.
- Online Masters Students:
 - Manuela Pop. Thesis Project: To be named - Neural Architecture Search (Current)
 - Phil Tinn. Thesis Project: Cross-Domain Adaptation and Geometric Data Synthesis for Near-Eye to Remote Gaze Tracking
- Other graduate research mentoring:
 - Balmukund Sharma. Master’s research assisting Christina Petlowany’s gaze/social-navigation work. Spring–Fall 2026.
 - Bruno Bertolo. MFA research on robot affect. Fall 2024–Fall 2025.
- I have also worked closely with several of Peter Stone’s and Luis Sentis’s students, as reflected in my publication record.

Master’s Reader

- John Lyle. Thesis: *Using Eye Gaze and Head Pose to Understand Behavior and Attention of Mobile Robot Operators*. Master of Science in Engineering, The University of Texas at Austin. Supervisor: Mitch Pryor. May 2026.
- Rolando Fernandez. Thesis: *Light-Based Nonverbal Signaling with Passive Demonstration for Mobile Service Robots*. With: Peter Stone. 2018.

Undergraduate Mentoring

UNDERGRADUATE THESIS SUPERVISION

- Victoria Zhou. UT CS Honors Thesis. “Exploration of Neural Networks for Stereo Vision.” 2019.
- Nick Walker. Polymathic Scholars Thesis: “An Interaction Perspective on Learning for Domestic Service Robots.” Bridging Disciplines Program Creative Project: “Exploring New Techniques for Generative Art.” 2018.

UNDERGRADUATE THESIS READER

- Tapaswini Kodavanti. Second reader, Turing Scholars honors thesis: *NeuroRule: Rule-Based Neural Network Distillation*. Computer Science, The University of Texas at Austin. Adviser: Risto Miikkulainen. May 2026. Defense: April 30, 2026.

- Harshal Viral Bharatia. Second reader, Turing Scholars honors thesis: *OraPlan: Oracle Assisted Hierarchical Planning and Learning with LLMs*. Computer Science, The University of Texas at Austin. Adviser: Peter Stone. November 2025. Defense: November 17, 2025.

OTHER UNDERGRADUATE SUPERVISION

The following 25 undergraduate students have performed research under my supervision that resulted in publications at UT Austin: Abrar Anwar, Akash Singh, Asha Jain, Blake Holman, Bonnie Mahajan, Connor Sheehan, Daksh Dua, Geethika Hemkumar, Gilbert Briscoe-Martinez, Harshavardhan Jagannathan, Henry Liu, Jacqueline Deans, Jamin Goo, Kathryn Baldauf, Mahathi Chillaria, Marika Murphy, Nathan John, Parth Chonkar, Rishi Shah, Nicholas Walker, Rolando Fernandez, Sean Kirmani, Shikhar Gupta, Stone Tejeda, Sydney Owen.

Additional students I have supervised on research prepared for submission or under review include Alekhya Kuchimanchi, David Tang, Efren Mendoza Enriquez, Hanaa Siddiqui, Henry Plante, Justin Medich, Mrityunjay (Manas) Mishra, Ruchi Bhalani, Saarthak Mohan, Sheryl Dsouza, Tiffany Kao.

The following 54 students have served as peer mentors (undergraduate TAs) for my & Peter's FRI stream: Abrar Anwar, Akash Singh, Alekhya Kuchimanchi, Anna Wang, Anwesha Roy, Aryan Behmardisharifabad, Athulya Saravanakumar, Bhavana Katta, Blake Holman, Bonnie Mahajan, Brigitte Krause, Connor Sheehan, Daksh Dua, David Tang, Eesha Kandukuri, Ethan Brown, Geethika Hemkumar, Hanaa Siddiqui, Harshal Bharatia, Henry Liu, Jacqueline Deans, Jamin Goo, Jeffrey Huang, Jennifer Suriadinata, Jonathan Browne, Joseph Moyalan, Kathryn Baldauf, Lucas Helms, Lucinda Nguyen, Madeleine Williams, Mahathi Chillaria, Marika Murphy, Matthew Kozlowski, Mayuri Raja, Mrityunjay (Manas) Mishra, Mukund Maini, Nalin Mahajan, Nathan John, Navya Agrawal, Nicholas Walker, Parth Chonkar, Pranav Rayudu, Rishi Shah, Ruchi Bhalani, Sanika Misra, Shikhar Gupta, Stone Tejeda, Sydney Owen, Vanna Chen, Victoria Zhou, Vijay Vuyyuru, Vineeth Bandi, Wyatt Hansen, Xiangwei (David) Chen.

I am also fortunate to have informally mentored 16 students during my time at both UBC and Yale. UBC Masters: Sara Shiekhleslami, Vidar Skjervøy (on exchange from University of Glasgow). UBC undergraduates: Vedanshu Dash, Louisa Hardjasa, Henry Mak, Alexander Toews. Yale Undergraduates: Ashley Douglas, David Golub, Elaine Short, Eleanor Avrunin, Gabriel Fernandez, Graham Radman, Justin Kosslyn, Kenny Casteneda, Michelle Vu, Wilma Bainbridge.

Funding

NSF Science and Technology Center, Award #2535195 *Center for Human and Robot Co-Adaptation*

Role: Senior Personnel; UT Austin HERO Facility Lead

Amount: \$30,000,000 (five-year, center-wide award)

Award announced: August 27, 2026.

UT Austin Portugal Program *EXPLAnation and ObseRvation for HumanOid DOMestic ROBots (EXPLOR-HODOR)*

2025 Call for Exploratory Research Projects; project 2025.17632.UTA

Role: Co-PI

Award allocation: Up to \$100,000 (UT Austin, including indirect costs)

Period: August 15, 2026–August 31, 2027.

UT Austin Portugal Program *FOundation MOdels for HumanOid DOmestic RObots*

Modelos Fundacionais para Robôs Domésticos Humanóides

2024 Call for Exploratory Research Projects under the UT Austin Portugal Program

Role: Co-PI

Period: September 2025 - September 2026.

Texas Robotics Industrial Affiliate Program

Vision-Language-Action (VLA) Models for Plan, Activity, and Intent Recognition and Multi-Agent Reasoning and Planning

Principal Investigators: Blake Anderson, Justin Hart, and Mitch Pryor

Award amount: \$25,000.

Texas Robotics Industrial Affiliate Program

A Foundation Model to Predict Whole-Body Behavior During Mobility with Robots — Year 2

Principal Investigators: Justin Hart and Nicholas Fey

Award amount: \$50,000.

Texas Robotics Industrial Affiliate Program

A Foundation Model to Predict Whole-Body Behavior During Mobility with Robots

Investigators: Justin Hart and Nicholas Fey

Award amount: \$50,000.

NSF Award #2219236 *GCR: Community-Embedded Robotics: Understanding Sociotechnical Interactions with Long-term Autonomous Deployments*

Role: Co-PI

Period: October 2022 - September 2024.

UT Austin Bridging-Barriers: Good Systems *Living and Working with Robots*

Role: Co-PI

Period: September 2021 - September 2026.

NSF Award #2125858 *NRT-AI: Convergent, Responsible, and Ethical Artificial Intelligence Training Experience for Roboticists*

Role: Co-PI

Period: September 2021 - August 2026.

Army Futures Command Robotics Center of Excellence *Persistent Fully Autonomous Multi-Robot Tactics in Complex Environments*

Role: Co-PI

Period: October 2019 - December 2022.

Cisco Systems *Semantic SLAM for Augmented Reality and Robotics*

Role: PI

Period: September 2022 - December 2024.